Multi-Scale Structural Descriptors for Governance-Relevant Patterns in Data Lineage Graphs

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Abstract

Data governance maturity is typically assessed through surveys, audits, and capability-model checklists that capture stated policy but not realized practice. We propose four multi-scale structural descriptors that detect governance-relevant patterns in the topology of data lineage graphs: community stability under resolution sweep (D1), blast-radius concentration profile (D2), spectral gap and Fiedler structure (D3), and persistent homology of shortest-path filtrations (D4). In controlled experiments on synthetic lineage graphs at three governance levels, D1, D2, and D4 achieve statistically significant differentiation after Benjamini–Hochberg FDR correction at the largest scale tested ($N \approx 130$), with Cohen's d exceeding 3.9 for D1 and 15.0 for D4. Validation on an anonymized production dbt lineage graph (223 nodes, 263 edges, 26 domains) yields permutation Spearman $\rho = -0.71$ between D3 algebraic connectivity and documentation rate across 18 domains (FDR-adjusted $p = 0.043$). Subset robustness analysis shows this association does not survive restriction to the 6 domains with internal edges ($p = 0.19$), and layer-stratified permutation confirms D3 is constant within each layer stratum ($p = 1.0$): the correlation is a between-layer architectural pattern, not a within-layer governance signal. A degree-preserving rewiring null model confirms that D1, D3 Fiedler bimodality, and D4 H1 features capture structure beyond degree sequences ($|z| > 2.9$); layer-preserving rewiring yields D1 $z = +2.68$ and D3 alg. conn. $z = -2.27$. Cross-dataset comparison on 32 graphs from four sources (WfCommons scientific workflows, DLG-DG-23 Huawei Cloud lineage graphs, DW-Bench warehouse schemas, and the dbt manifest) shows that production data lineage graphs from two independent organizations converge on high community stability and small spectral gaps, occupying a region of descriptor space distinct from scientific workflows. These results suggest that lineage topology encodes governance-relevant structural signal, though the governance correlation analysis rests on a single case study and generalization requires further validation.

Keywords: data governance, lineage graphs, community detection, spectral graph theory, persistent homology, structural descriptors, network analysis

1 Introduction

Data governance frameworks specify how organizations should manage data assets: who owns them, how quality is assured, what documentation exists, and how changes propagate through transformation pipelines. Mature governance manifests as clear domain boundaries, tested transformations, and documented interfaces between stages of a data pipeline. Immature governance manifests as orphaned assets, unclear ownership, redundant paths, and untested transformations.

The dominant approach to assessing governance maturity relies on self-assessment against capability models such as DAMA-DMBOK [DAMA International, 2017] or DCAM. These frameworks are reference guides: they organize knowledge areas and suggest best practices, but do not prescribe a single maturity ladder. More fundamentally, they capture stated policy rather than the actual structural organization of data assets. Two organizations with similar self-assessment scores can have very different lineage topologies: one with clean layered DAGs and well-separated domain boundaries, another with tangled cross-domain dependencies and redundant paths.

We hypothesize that governance practices leave structural fingerprints in data lineage graphs. If domain boundaries are enforced, community detection at the appropriate resolution should recover stable, modular partitions. If transformations are tested, the blast radius of any single failure should be concentrated rather than diffuse. If the pipeline architecture is clean, the spectral properties of the lineage graph should reflect appropriate coupling. If redundant dependency paths exist (a sign of organic growth without architectural oversight), persistent homology should detect reconvergent structure in the undirected skeleton.

This paper introduces four multi-scale structural descriptors designed to capture these properties, validates them on synthetic lineage graphs with controlled governance variation, tests their sensitivity to governance-relevant interventions, and provides initial validation on a real production dbt lineage graph with genuine governance metadata.

1.1 Contributions

1. Four graph-theoretic descriptors (D1–D4) that detect governance-relevant structural patterns from lineage topology, each operating at a different structural scale.

2. An empirical validation framework with synthetic lineage generators at three governance levels, demonstrating statistically significant differentiation with FDR correction and a degree-preserving null model.

3. Initial real-data validation on an anonymized production dbt lineage graph (223 nodes, 26 domains) with permutation-based inference, partial correlations controlling for structural confounds, and a cycle-rank ablation for the topological descriptor.

2 Related Work

2.1 Governance Measurement

Data governance frameworks define structural and procedural mechanisms for data management [Khatri and Brown, 2010, Abraham et al., 2019]. Assessment typically relies on maturity models: DAMA-DMBOK [DAMA International, 2017] organizes eleven knowledge areas as a reference guide (not a prescriptive maturity ladder); DCAM evaluates institutional readiness. Weber et al. [2009] argue that governance design is contingent on organizational strategy, implying that a single maturity scale is insufficient.

Institutional theory provides a theoretical lens. DiMaggio and Powell [1983] describe three isomorphic pressures (coercive, mimetic, normative) that cause organizations to converge structurally under shared governance regimes. Scott [2013] extends this via regulative, normative, and cultural-cognitive pillars. We adopt this view: if governance is an institutional arrangement, it should produce detectable structural signatures in the artifacts it governs.

2.2 Graph Analysis of Data Pipelines

Hou et al. [2026] apply temporal convolutional and graph convolutional networks to data lineage DAGs for fault classification (1,240 nodes, 5,870 edges), achieving 95.8% accuracy. Their work uses lineage structure as input to prediction but does not extract governance-relevant patterns from that structure.

Knowledge graph quality metrics [Seo et al., 2022] define six structural measures for knowledge graphs. Meroño Peñuela et al. [2024] position knowledge graphs as governance backbones for AI systems. Neither applies multi-scale graph descriptors to lineage topology.

Preliminary work on fractal descriptors for enterprise data graphs [Hari, 2026b] investigated whether box-covering fractal dimension could characterize governance-relevant structure in data lineage. That investigation found fractal dimension to be sensitive to graph scale and topology but insufficient as a standalone governance signal, motivating the broader multi-scale descriptor set developed here. The fractal analysis methodology (Hurst exponents, detrended fluctuation analysis) has a longer history in the author's work on temporal scaling in financial systems [Hari, 2026c]. A complementary line of work applies boundary-risk detection to master data governance [Hari, 2026a], identifying entity clusters whose governance status is sensitive to local threshold perturbation. That approach targets record-level entity resolution rather than pipeline-level lineage topology, but shares the premise that governance properties are recoverable from structural analysis of data artifacts.

2.3 Community Detection at Multiple Resolutions

The resolution parameter in modularity optimization [Reichardt and Bornholdt, 2006] controls community granularity. Lambiotte et al. [2014] interpret the parameter as Markov diffusion time. Delvenne et al. [2010] introduce stability as plateau detection via normalized variation of information (NVI). The Leiden algorithm [Traag et al., 2019] guarantees connected communities, addressing a known Louvain failure mode. Our implementation uses the Louvain algorithm (via NetworkX) rather than Leiden, as the resolution sweep measures partition stability across resolutions rather than depending on within-resolution community connectedness guarantees.

2.4 Spectral Methods

Algebraic connectivity (the second-smallest Laplacian eigenvalue) measures graph cohesion [Fiedler, 1973]. Spectral gap correlates with robustness to random failure and inversely with mean path length. Von Neumann spectral entropy [De Domenico and Biamonte, 2016] quantifies complexity of the eigenvalue distribution. We compose these into a descriptor that captures coupling structure between pipeline components.

2.5 Persistent Homology on Networks

Topological data analysis via persistent homology has been applied to brain networks, social networks, and protein interaction networks [Otter et al., 2017, Petri et al., 2013]. Chowdhury and Mémoli [2018] introduce directed persistent path homology. For DAGs, directed H1 is trivially zero (no directed cycles by definition), but H1 in the undirected skeleton detects reconvergent dependency paths where multiple routes exist between a source and its consumers. To our knowledge, no prior work applies persistent homology to data lineage or governance graphs.

3 Descriptors

We define four structural descriptors for directed acyclic data lineage graphs $G = (V, E)$ with $|V| = N$ nodes and $|E| = M$ edges. Each node represents a data asset (source, transformation, or mart). Edges represent dependency: $(u, v) \in E$ means v depends on u.

Graph preprocessing. Production lineage graphs are often weakly disconnected: isolated source tables and small clusters that share no edges with the main pipeline appear in every real dataset we examined. Descriptors D1, D3, and D4 are computed on the largest weakly connected component (LWCC) of the undirected projection of G, since community detection and spectral methods require a connected graph. D2 (blast-radius) uses the full directed graph because downstream reachability is defined per node regardless of global connectivity. For the production dbt graph, the LWCC contains 185 of 223 nodes (83%); the 38 remaining nodes are isolated source tables with no intra-domain edges.

3.1 D1: Community Stability Index (CSI)

We sweep the resolution parameter γ of the Louvain algorithm over $[0.1, 10.0]$ in 20 logarithmically spaced steps on the undirected skeleton of G, computing community partitions $\mathcal{C}(\gamma)$ at each resolution. For adjacent resolution pairs (γ_i, γ_{i+1}), we compute the normalized variation of information:

$$\mathrm{NVI}(\gamma_i, \gamma_{i+1}) = \frac{\mathrm{VI}(\mathcal{C}_i, \mathcal{C}_{i+1})}{\log_2 N} \tag{1}$$

where $\mathrm{VI} = H(\mathcal{C}_i|\mathcal{C}_{i+1}) + H(\mathcal{C}_{i+1}|\mathcal{C}_i)$ is the variation of information and $\log_2 N$ is the maximum possible entropy for N nodes. The Community Stability Index counts the fraction of consecutive resolution steps where the partition remains stable:

$$\mathrm{CSI} = \frac{|\{i : \mathrm{NVI}(\gamma_i, \gamma_{i+1}) < \tau\}|}{n_{\mathrm{steps}} - 1}, \quad \tau = 0.1 \tag{2}$$

CSI $\in [0, 1]$, with higher values indicating more stable community structure across resolutions. We additionally extract fragmentation onset (the γ at which the number of communities first exceeds twice the minimum), modularity at $\gamma = 1$, and number of communities at $\gamma = 1$.

Louvain stochasticity. The Louvain algorithm is a heuristic with random seed dependence. We audited D1 CSI over 25 random seeds per graph and report seed-NVI at $\gamma \approx 1.0$ (mean pairwise NVI between seed partitions at a fixed resolution) as a measure of optimizer variance. $\mathrm{CSI}_{\mathrm{std}} < 0.05$ and seed-NVI < 0.05 indicates the result is not an artifact of seed choice.

Governance interpretation. Well-governed pipelines with clear domain boundaries should exhibit high CSI: communities are robust across resolutions because they reflect genuine organizational structure. Poorly governed pipelines with tangled cross-domain dependencies should exhibit low CSI.

3.2 D2: Blast-Radius Concentration Profile

For each node $v \in V$ and depth $d \in \{1, 2, \ldots\}$, we compute $R_d(v) = |\{u : \mathrm{dist}(v, u) = d\}|$, the number of nodes reachable at exactly depth d downstream. The Gini coefficient of $\{R_d(v)\}_{v \in V}$ at each depth d yields a concentration profile:

$$G(d) = \mathrm{Gini}(\{R_d(v) : v \in V\}) \tag{3}$$

We extract the maximum Gini across depths and the concentration transition depth (the depth at which Gini first exceeds 0.5, or -1 if never reached).

Governance interpretation. In well-governed pipelines, blast radius should be concentrated: a few hub nodes affect many downstream assets, while most nodes have limited impact. High max Gini indicates concentrated risk, which, when combined with stewardship data, can reflect intentional architectural design.

3.3 D3: Spectral Gap and Fiedler Structure

From the combinatorial (unnormalized) Laplacian $L = D - A$ of the undirected skeleton, where D is the degree matrix and A the adjacency matrix, we extract:

- Algebraic connectivity λ_2: the second-smallest eigenvalue of L on the LWCC. Because the LWCC is connected by construction, $\lambda_1 = 0$ and $\lambda_2 > 0$ always.
- Normalized spectral gap: $\lambda_2/\lambda_{\max}$, where $\lambda_{\max}$ is the largest eigenvalue.
- Fiedler bimodality: bimodality coefficient of the Fiedler vector (the eigenvector associated with λ_2).
- Spectral entropy: von Neumann entropy of the eigenvalue distribution, $S = -\sum_i \bar{\lambda}_i \log_2 \bar{\lambda}_i$ where $\bar{\lambda}_i = \lambda_i / \sum_j \lambda_j$ (computed over nonzero eigenvalues).

Governance interpretation. Algebraic connectivity measures overall graph cohesion. Low values indicate loosely connected components (possibly orphaned domains). High Fiedler bimodality indicates a natural bipartition, potentially reflecting a two-tier architecture (e.g., source vs. mart layers). Spectral entropy captures the complexity of the coupling structure.

3.4 D4: Persistent Homology via Shortest-Path Filtration

We compute the all-pairs shortest-path distance matrix on the undirected skeleton of G and construct a Vietoris–Rips filtration. For each filtration threshold ϵ, we compute homology groups in dimensions 0 and 1. The resulting persistence diagram records birth and death times for each topological feature.

From the H1 (cycle) persistence diagram, we extract:

- Number of H1 bars (n_{H1}) and its normalization n_{H1}/N.
- Total H1 persistence: $P_1 = \sum_i (d_i - b_i)$ over all H1 bars.
- H1 persistence entropy: $S_1 = -\sum_i p_i \log_2 p_i$ where $p_i = (d_i - b_i)/P_1$.

As a simpler baseline, we also compute the cycle rank $\beta_1 = M - N + C$ of the undirected skeleton (where C is the number of connected components), which counts the total number of independent cycles without regard to their persistence or scale.

Governance interpretation. H1 features in the undirected skeleton represent reconvergent dependency paths: alternative routes between a source node and its consumers that do not exist in a clean tree-structured pipeline. Unlike cycle rank, persistent homology distinguishes between long-lived structural cycles (reflecting persistent architectural redundancy) and short-lived transient features. High n_{H1}/N indicates reconvergent dependency structure.

4 Experimental Design

4.1 Synthetic Lineage Generator

We generate directed acyclic lineage graphs at four scales (tiny: $N \approx$ 14–34; small: $N \approx$ 28–48; medium: $N \approx$ 65–81; large: $N \approx$ 114–133) and three governance quality levels:

Well-governed: Three clearly separated domains with layered structure (source $\to$ staging $\to$ mart). Cross-domain edges only at mart layer. Test and documentation coverage assigned to 70–90% of nodes.

Baseline: Same domain count but with moderate cross-domain coupling at all layers. Test coverage 40–60%.

Poorly governed: Flat structure with frequent cross-domain shortcuts and orphaned nodes. Test coverage 10–30%. Random edge additions creating shortcut paths.

Each condition is replicated with 10 random seeds, yielding $4 \times 3 \times 10 = 120$ synthetic graphs.

4.2 Experiment 1: Governance Differentiation

For each graph, we compute all four descriptors (D1–D4) plus cycle rank, nine single-scale baselines (degree, betweenness, PageRank statistics; diameter; density), and three MTTD-based outcome measures. Mann–Whitney U tests compare well-governed vs. poorly governed at each scale, with Benjamini–Hochberg FDR correction applied across all tests within each scale.

4.3 Experiment 2: Real Data Validation

We validate on an anonymized production dbt manifest lineage (223 nodes, 263 edges, 26 domains) exported using HMAC-SHA256 anonymization. This graph includes genuine governance metadata: test counts, documentation flags, and layer assignments derived from the dbt catalog.

For the dbt lineage, we partition by domain and compute permutation-based Spearman correlations (10,000 permutations) between per-domain descriptor values and governance metadata (test rate, documentation rate, composite governance score). All p-values are FDR-corrected via Benjamini–Hochberg. For D3 descriptors, we additionally report partial Spearman correlations controlling for domain size, edge count, and source fraction to address size-related confounding.

4.4 Experiment 2b: Degree-Preserving Null Model

To test whether the descriptors capture structure beyond what degree sequences alone determine, we apply 100 degree-preserving directed edge rewirings to the real dbt graph ($10M$ swaps per rewiring). We compute D1–D4 and cycle rank on each rewired graph and report z-scores of the real values relative to the null distribution.

4.5 Experiment 3: Intervention Sensitivity

We apply four governance-relevant interventions to a baseline synthetic graph ($N = 132$, 3 domains): domain reorganization, orphan removal, shortcut removal, and monitor placement. Each intervention is compared against 60 random perturbation trials at the same magnitude. The signal-to-noise ratio (SNR) is the intervention effect size divided by the standard deviation of random perturbation effects.

4.6 Experiment 5: Cross-Dataset Structural Characterization

To test whether the descriptors generalize beyond our synthetic generators and single dbt instance, we compute D1–D4 on three open datasets: 11 scientific workflow DAGs from WfCommons [Coleman et al., 2022] spanning astronomy, bioinformatics, and agroecosystem applications (101–4,846 nodes); 18 production data lineage graphs from DLG-DG-23 [Chen et al., 2024], extracted from Huawei Cloud DataArts governance projects (24–1,059 table-level nodes after restricting to Data Table and Data Job assets with DATA_FLOW edges); and 2 data warehouse schemas from DW-Bench [Ahmed and Sakar, 2026] (OMOP clinical data model, TPC-DI brokerage; 31–37 tables). These lack the domain-level governance metadata needed for correlation analysis, so the comparison is purely structural: do real pipeline architectures from different domains and organizations occupy distinguishable regions of descriptor space?

4.7 Experiment 4: Predictive Comparison

We compare multi-scale features (D1–D4) against single-scale baselines in two prediction tasks: classification of governance level (logistic regression and random forest) and regression of Mean-Time-to-Detect (Ridge and random forest). All models are evaluated via 5-fold cross-validation.

Note on MTTD. The MTTD simulation places monitors at nodes flagged as stewards or at high-betweenness nodes. Since stewardship data overlaps with governance metadata, MTTD is not independent of the governance labels used elsewhere. The regression task tests whether structural descriptors predict MTTD beyond what single-scale graph statistics capture, not whether MTTD independently validates the descriptors.

5 Results

5.1 Experiment 1: Governance Differentiation

Table 1: Governance differentiation at large scale ($N \approx$ 114–133). Mann–Whitney U test, well-governed vs. poorly governed, 10 replicates per condition. All p-values BH FDR-corrected.

Descriptor	Well	Poor	d	U	raw p	FDR p
D1 CSI	0.653	0.374	+3.90	100	< 0.001	< 0.001
D1 frag. onset	0.800	0.154	+4.74	100	< 0.001	< 0.001
D2 max Gini	0.696	0.678	+1.71	90	0.003	0.005
D3 Fiedler bim.	0.709	0.864	−1.22	14	0.007	0.011
D3 norm. gap	0.005	0.005	−0.11	55	0.734	0.769
D4 H1/N	0.589	0.743	−16.0	0	< 0.001	< 0.001
D4 cycle rank/N	0.589	0.743	−16.0	0	< 0.001	< 0.001

After FDR correction, D1 (CSI, fragmentation onset), D2 (max Gini), D3 (Fiedler bimodality), and D4 (H1/N) achieve statistically significant differentiation at the large scale. D3 normalized spectral gap does *not* survive FDR correction (FDR $p = 0.77$), a discrepancy we revisit in the null model analysis.

D4 (H1/N) shows the largest effect size ($|d| = 16.0$): poorly governed graphs have more reconvergent dependency paths per node. D1 CSI and fragmentation onset show large effects ($d > 3.9$), confirming that domain boundary quality is recoverable from community structure. D2 max Gini shows the weakest differentiation ($d = 1.7$).

D4 ablation. At the large scale, H1/N and cycle rank/N yield identical effect sizes ($d = -16.0$) because the synthetic generators produce graphs where all H1 cycles have similar persistence. At the tiny scale, a modest divergence appears (H1/N $d = -11.5$ vs. cycle rank/N $d = -9.0$). In this synthetic setting, the simpler cycle rank captures the same governance signal as persistent homology.

5.2 Experiment 2: Real Data Validation

5.2.1 Production dbt Lineage

The anonymized dbt lineage graph (223 nodes, 263 edges) is a verified DAG with three layers (155 source/raw, 33 silver/intermediate, 35 gold/mart) and 26 anonymized domains. The largest weakly connected component contains 185 nodes and 262 edges.

Governance metadata reveals a clear gradient: gold/mart nodes have 86% test coverage and 91% documentation, while source/raw nodes have 0% test coverage and 41% documentation. Silver/intermediate nodes have 0% for both. No steward metadata was available.

After FDR correction, D3 algebraic connectivity and normalized gap are associated with documentation rate (FDR $p = 0.043$) and marginally with composite governance score (FDR $p = 0.080$). No other descriptors survive FDR correction.

5.2.2 D4 and Test Rate: A Qualitative Observation

The $\rho = 1.0$ correlation between D4 H1/N and test rate is driven by a binary split: exactly one domain (domain_012, the gold/mart layer, 35 nodes, 63 edges) has both H1 topological

Table 2: Permutation Spearman correlations between per-domain descriptors and governance metadata (18 domains with $N \geq 5$). Permutation p-values from 10,000 permutations; FDR p via Benjamini–Hochberg.

Descriptor vs. target	ρ	perm. p	FDR p	
vs. composite governance score				
D3 algebraic connectivity	−0.643	0.006	0.080	†
D3 normalized gap	−0.635	0.007	0.080	†
D3 spectral entropy	−0.447	0.067	0.123	
vs. documentation rate				
D3 algebraic connectivity	−0.708	0.002	0.043	*
D3 normalized gap	−0.701	0.002	0.043	*
D3 Fiedler bimodality	−0.561	0.017	0.116	
D2 max Gini	−0.563	0.017	0.116	
vs. test rate (qualitative only; see text)				
D4 H1/N	+1.000	0.058	0.116	
D4 cycle rank/N	+1.000	0.058	0.116	

† FDR < 0.10; * FDR < 0.05

features and nonzero test coverage. The remaining 17 analyzed domains have zero for both. Under permutation, the probability of this alignment by chance is approximately $1/18 \approx 0.056$. This represents a case-study observation consistent with topological complexity and testing co-occurring in a production system. It is not evidence of a continuous relationship, and we do not claim it as a confirmed finding.

5.2.3 Partial Correlations

Because D3 descriptors may correlate with governance metadata simply because larger, denser domains tend to have different governance practices, we computed partial Spearman correlations controlling for domain size (N), edge count (M), and source fraction.

Table 3: D3 partial correlations controlling for domain size, edge count, and source fraction (18 domains, permutation p from 10,000 perms).

D3 descriptor	Target	Raw ρ	Partial ρ	Partial p
Algebraic conn.	governance score	−0.643	−0.684	0.002
Algebraic conn.	documentation rate	−0.708	−0.546	0.021
Norm. gap	governance score	−0.635	+0.171	0.488
Norm. gap	documentation rate	−0.701	+0.113	0.650
Fiedler bim.	governance score	−0.446	+0.451	0.061
Spectral entropy	governance score	−0.447	+0.060	0.809

Only D3 algebraic connectivity survives partial correlation: the association with governance score *strengthens* slightly (partial $\rho = -0.68$, $p = 0.002$) and remains significant for documentation rate (partial $\rho = -0.55$, $p = 0.021$). The normalized spectral gap, despite having the second-strongest raw correlation, reverses sign after controlling for confounds and becomes nonsignificant. This indicates that the normalized gap association was confounded by domain size.

5.2.4 Cycle Rank Ablation

In the real-data analysis, H1/N and cycle rank/N produce identical correlations with all governance targets ($\rho = +0.404$ vs. governance score for both, $p = 0.058$). On this dataset, the simpler cycle rank captures the same signal as persistent homology, likely because the single structurally complex domain dominates both measures.

5.2.5 Metadata-Only Negative Control

Of 18 domains with $N \geq 5$, twelve are source-only clusters with zero internal edges. These domains have nonzero governance metadata (documentation rates from 0.04 to 1.00) but degenerate descriptor values (all D1–D4 descriptors equal zero). If the descriptors were merely recapitulating governance metadata through some hidden correlation, these domains would show descriptor variation. Their uniform descriptor values confirm that the descriptors respond to structural topology, not metadata artifacts.

5.3 Experiment 2b: Null Model

Table 4: Z-scores of real dbt graph descriptors relative to 100 degree-preserving rewirings.

Descriptor	Real	Null mean	Null std	z
D1 CSI	0.947	0.615	0.090	+3.70***
D1 n communities	10	13.2	1.02	−3.14***
D2 max Gini	0.480	0.579	0.042	−2.33**
D3 alg. connectivity	0.067	0.071	0.011	−0.42
D3 norm. gap	0.003	0.003	0.001	−0.38
D3 Fiedler bimodality	0.302	0.934	0.027	−23.8***
D3 spectral entropy	6.674	6.679	0.003	−1.64
D4 H1/N	0.220	0.272	0.018	−2.95**
D4 H1 entropy	5.563	5.910	0.097	−3.57***
D4 cycle rank/N	0.350	0.347	0.003	+0.88

The null model yields three classes of results:

Structure beyond degree sequence. D1 CSI ($z = +3.70$), D3 Fiedler bimodality ($z = -23.8$), and D4 H1 features ($z = -2.95$ to -3.57) are significantly different from degree-preserving rewirings. The real graph has higher community stability, lower Fiedler bimodality, and fewer H1 cycles than degree-matched random graphs. These descriptors capture genuine structural organization.

Degree-determined descriptors. D3 algebraic connectivity ($z = -0.42$) and normalized spectral gap ($z = -0.38$) are indistinguishable from the null. Their values are largely determined by the degree sequence, not by governance-relevant topology. This is consistent with the partial correlation results: normalized gap lost significance after controlling for size-related confounds.

Cycle rank. As expected, cycle rank ($z = +0.88$) varies minimally under degree-preserving rewiring because M and N are fixed, and the number of connected components changes only slightly. In contrast, H1 features change substantially ($z = -2.95$), confirming that persistent homology captures structural information that cycle rank does not.

5.4 Experiment 2c: D1 Seed Robustness

Louvain is stochastic. We ran the D1 resolution sweep with 25 random seeds on the dbt manifest, two synthetic graphs ($N = 130$), and 15 DLG-DG-23 graphs ($N \leq 400$). For the dbt manifest, CSI $= 0.87 \pm 0.08$ (range 0.68–1.00, seed-NVI $= 0.062$). CSI is consistently high across seeds—the minimum over 25 seeds is still 0.68—confirming a genuine signal. Synthetic graphs show higher variance (std 0.09–0.14), and DLG-DG-23 graphs are generally stable (seed-NVI < 0.07 for 11/15 graphs, mean CSI $= 0.85 \pm 0.06$). The three DLG graphs with seed-NVI > 0.10 are the largest ones ($N > 100$), where Louvain optimizer noise increases.

5.5 Experiment 2d: Extended Null Models

Table 5: Z-scores of real dbt descriptors relative to 100 layer-preserving DAG rewirings (swap acceptance rate 22.8%; 11% of rewired graphs remain DAGs).

Descriptor	Real	Null mean	Null std	z
D1 CSI	0.947	0.651	0.111	+2.68**
D3 alg. connectivity	0.067	0.114	0.021	−2.27**
D3 norm. gap	0.003	0.005	0.001	−2.28**
D3 Fiedler bimodality	0.302	0.624	0.216	−1.50
D2 max Gini	0.480	0.496	0.021	−0.79
D4 cycle rank/N	0.350	0.346	0.004	+0.91

Layer-preserving rewiring constrains swaps to edge pairs where source layers match and target layers match, accepting 22.8% of attempts. Only 11% of rewired graphs remain DAGs, so this null is a conservative stress test rather than a proper lineage null. D1 ($z = +2.68$) and D3 algebraic connectivity ($z = -2.27$) remain significant. D3 Fiedler bimodality does not ($z = -1.50$), inconsistent with the unconstrained null ($z = -23.8$); this discrepancy reflects the larger variance of the layer-constrained null distribution (std 0.216 vs. 0.027), likely because the low swap acceptance rate limits structural diversity among rewired graphs.

5.6 Experiment 2e: Subset Robustness and Layer Confound in D3

Table 6: Subset robustness for D3 algebraic connectivity vs. documentation rate. Internal edges: $M_{\text{internal}} > 0$.

Subset	n	ρ	FDR p	Status
A: all $N \geq 5$	18	−0.708	0.029	Survives
B: internal edges only	6	−1.000	0.189	Fails
C: no source-dominant	6	−1.000	0.189	Fails
D: no largest domain	17	−0.806	< 0.01	Spurious†

†Significance driven by test_rate constant-input artifact.

D3 takes the value 0 for all 12 source-dominant domains (no internal edges) and the value 1.0 for all 5 silver/intermediate domains (small fully-connected clusters), with intermediate D3 for the gold/mart domain. Layer-stratified permutation (5,000 permutations of documentation labels within each layer stratum) yields $p = 1.000$: D3 is constant within each stratum, so every permutation reproduces the real correlation exactly. The D3–documentation association is a between-layer architectural pattern. It does not capture within-layer governance quality.

5.7 Experiment 3: Intervention Sensitivity

Each descriptor responds most strongly to a different intervention type. D4 is most sensitive to domain reorganization (SNR = 8.67), D3 normalized gap to orphan removal (SNR = 6.53),

Table 7: Signal-to-noise ratios for governance interventions. SNR > 2 indicates the descriptor reliably detects the intervention above random perturbation noise.

Descriptor	Add domains	Remove orphans	Remove shortcuts	Add monitors
D1 CSI	1.43	1.18	0.12	2.19
D1 frag. onset	3.25	3.87	0.08	0.41
D2 max Gini	1.87	0.93	0.56	2.79
D3 norm. gap	4.21	6.53	0.31	1.15
D3 entropy	6.85	2.74	0.22	0.88
D4 H1/N	8.67	1.96	0.14	1.33

and D2 to monitor placement (SNR = 2.79). Shortcut removal produces weak signal across all descriptors because the intervention magnitude (4 edges removed from a 132-node graph) falls below the noise floor.

5.8 Experiment 5: Cross-Dataset Structural Characterization

To assess whether real pipeline topologies occupy a distinct region of descriptor space, we computed D1–D4 on 11 scientific workflow DAGs from WfCommons [Coleman et al., 2022] (101–4,846 nodes), 18 production data lineage graphs from DLG-DG-23 [Chen et al., 2024] (24–1,059 table-level nodes), two data warehouse schemas from DW-Bench [Ahmed and Sakar, 2026] (31–37 tables), and the production dbt manifest.

Table 8: Structural profiles across 32 graphs from four sources. WfCommons and DLG-DG-23 values are means with ranges in parentheses.

Source	n	CSI	max Gini	gap	Fiedler bim.	CR/N
WfCommons (11)	101–4846	0.78	0.39	0.011	0.59	0.91
		(.53–.84)	(.00–.80)	(.00–.04)	(.28–.93)	(.00–2.99)
DLG-DG-23 (18)	24–1059	0.87	0.61	0.003	0.66	0.26
		(.68–1.0)	(.35–.73)	(.00–.01)	(.33–.98)	(.00–.68)
DW-Bench (2)	31–37	0.68	0.54	0.030	0.66	0.95
dbt manifest (1)	223	0.95	0.48	0.003	0.30	0.35

The DLG-DG-23 graphs, which represent production data lineage from Huawei Cloud governance projects, are the most structurally similar to the dbt manifest: both show high community stability (CSI $\geq$ 0.68, mean 0.87 vs. 0.95 for dbt), very small normalized spectral gaps ($<$ 0.014), and moderate cycle rank density (CR/N $\leq$ 0.68). Three DLG graphs achieve CSI = 1.0, indicating perfect community stability across the full resolution sweep. Scientific workflows, by contrast, show wider topological variation: seismology pipelines are nearly tree-structured (CR/N = 0), while bioinformatics alignment workflows (BWA, BLAST) have CR/N $>$ 1.9.

The production dbt manifest retains the lowest Fiedler bimodality (0.30) of all 32 graphs, suggesting the clearest spectral separation between its two dominant subgraphs. DW-Bench schemas occupy an intermediate position with moderate CSI but the highest spectral gaps ($>$ 0.01), reflecting their shorter dependency chains.

These results confirm that the descriptors produce interpretable variation across genuinely different pipeline architectures, not only synthetic governance variants. The convergence of dbt and DLG-DG-23 descriptor profiles, both from governed production environments but from different organizations and technology stacks, suggests that data lineage graphs exhibit structural regularities distinct from scientific workflows and data warehouse schemas. Alternative explanations (domain differences, scale effects) cannot be ruled out from this comparison alone, but the consistency across 19 independent production lineage graphs from two organizations is encouraging.

5.9 Experiment 4: Predictive Comparison

Table 9: Prediction performance (5-fold cross-validation, 120 samples).

Task	Feature set	Baselines	Multi-scale
Classification (AUC)	Logistic	1.000	1.000
Classification (AUC)	Random Forest	1.000	1.000
Regression R^2 (MTTD)	Ridge	0.246	0.277
Regression R^2 (MTTD)	Random Forest	0.175	0.307

Classification achieves ceiling AUC = 1.0 for all feature sets, confirming that synthetic governance variation produces structurally separable graphs but providing no discriminative evidence between feature sets. The informative comparison is MTTD regression: multi-scale features explain 75% more variance than baselines ($R^2 = 0.307$ vs. 0.175, random forest). Combined features do not improve over multi-scale alone, suggesting the multi-scale descriptors subsume the information in single-scale statistics for this prediction task.

6 Discussion

6.1 Which Descriptors Survive Scrutiny

The convergent evidence from synthetic experiments, real-data correlations, partial correlations, and null models identifies which descriptors capture governance-relevant structure and which do not.

D1 CSI passes all tests: it differentiates synthetic governance levels after FDR correction ($d = 3.9$), and the null model confirms it captures structure beyond degree sequences ($z = +3.7$). Its real-data correlation with governance metadata is not tested here because most domains lack internal edges for meaningful community detection.

D3 algebraic connectivity showed the strongest initial real-data correlation ($\rho = -0.71$ vs. documentation, FDR $p = 0.043$), but subset robustness analysis and layer-stratified permutation reveal this is a between-layer architectural artifact: D3 = 0 for source domains, D3 = 1.0 for silver domains, intermediate for gold/mart. The association does not survive restriction to domains with internal edges (Subset B, $n = 6$, FDR $p = 0.19$). The null model is consistent: D3 algebraic connectivity is degree-sequence-determined ($z = -0.42$), not capturing hidden structural signal. D3 characterizes layer architecture reliably, but cannot distinguish governance levels within layers.

D3 Fiedler bimodality is the most striking null model result ($z = -23.8$): the real graph has dramatically lower bimodality than degree-matched random graphs, suggesting that the real graph's Fiedler vector reflects genuine organizational structure (a smooth gradient across pipeline layers) rather than the sharp random bipartitions that emerge from rewiring.

D4 H1 features differentiate governance levels in synthetic experiments and capture structure beyond degree sequences ($z = -2.95$), while cycle rank does not ($z = +0.88$). This is the setting where persistent homology adds value over the simpler baseline: when testing whether structure goes beyond what degree sequences determine, cycle rank trivially passes (it is nearly preserved by rewiring), while H1 features provide a meaningful test.

D3 normalized spectral gap fails both the partial correlation test (sign reversal, $p = 0.49$) and the null model ($z = -0.38$). Its apparent real-data correlation was confounded by domain size.

6.2 The D3 Correlation Is Between-Layer, Not Within-Layer

The D3 algebraic connectivity–documentation association ($\rho = -0.71$, FDR $p = 0.043$) is a between-layer architectural pattern, not a within-layer governance signal. Source-dominant domains all have D3 = 0 (no internal edges) and varying documentation; silver/intermediate domains all have D3 = 1.0 (small fully-connected clusters) and zero documentation; the gold/mart domain has intermediate D3 and high documentation. D3 is constant within each layer stratum, which means layer-stratified permutation of governance labels cannot change the observed correlation: every permutation yields the same rho (empirical $p = 1.0$).

The correlation survives the full-sample permutation test (Subset A) because it captures real variation in architectural layer, not because it carries within-layer governance signal. Restricting to the six domains with internal edges (Subset B, $n = 6$) destroys power and the association fails FDR correction ($p = 0.19$). This is an honest negative result. D3 identifies layer boundaries in lineage graphs (a real structural signal), but the specific doc_rate correlation in this case study is confounded by layer composition rather than reflecting a direct governance-to-topology relationship.

6.3 Limitations

Limited governance correlation data. The governance correlation analysis (Experiment 2) rests on a single organization's dbt lineage with 18 analyzable domains. Experiment 5 extends structural characterization to 32 graphs across four sources, including 18 production lineage graphs from DLG-DG-23 [Chen et al., 2024]. The convergence of dbt and DLG-DG-23 profiles on high CSI and small spectral gaps is suggestive, but the DLG-DG-23 public release lacks domain-level governance metadata (documentation rate, test coverage, owner assignment), so we cannot test whether the descriptor–governance correlations replicate. Full replication requires an organization willing to share both lineage topology and governance metadata at domain granularity.

Small sample size. The 18 analyzable domains provide limited statistical power. Of these, only 6 have internal edges sufficient for meaningful descriptor computation. Larger or more architecturally complex lineage graphs would provide more per-domain variation.

Missing metadata. The production dbt catalog contains no steward or owner metadata, reducing the governance score to a two-component measure (tests + documentation). Organizations with richer metadata would enable testing D1's hypothesized association with ownership structure.

Synthetic generator validity. The governance variation in our synthetic generators is architectural (edge patterns, domain boundaries), not metadata-only. The generators produce cleaner separation between governance levels than likely exists in practice, contributing to the large effect sizes and classification ceiling.

Causal direction. Correlations between descriptors and governance metadata could reflect either direction: governance practices producing cleaner structure, or cleaner structure enabling easier governance. We measure association, not causation.

D4 cycle rank equivalence. In both the synthetic and real-data settings, cycle rank captures the same governance signal as H1 persistent features. The null model is the only test where H1 adds value over cycle rank. For practical applications, cycle rank may be sufficient and is vastly cheaper to compute.

6.4 Practical Implications

If validated on additional datasets, these descriptors could support:

1. **Structural monitoring.** Computing D1 and D3 Fiedler bimodality on each CI/CD push provides a structural signal without manual assessment. CSI degradation over time

signals erosion of domain boundaries.

2. **Targeted intervention.** D3 identifies structurally isolated components. D2 identifies nodes with outsized blast radius. These guide where to invest governance effort.

3. **Complementary evidence.** The descriptors complement existing governance frameworks by providing structural measurement where traditional methods provide periodic assessment.

7 Conclusion

We introduced four multi-scale structural descriptors for detecting governance-relevant patterns in data lineage graph topology. Synthetic experiments confirm that D1 (community stability), D2 (blast-radius concentration), and D4 (persistent homology) differentiate governance levels with statistical significance after FDR correction. A degree-preserving null model shows that D1, D3 Fiedler bimodality, and D4 H1 features capture structure beyond degree sequences. Real-data validation on a production dbt lineage graph yields a significant association between D3 algebraic connectivity and documentation rate that survives partial correlation controlling for structural confounds.

The results are drawn from a single case study and should be interpreted as evidence of feasibility, not generalization. D3 normalized spectral gap, which showed the second-strongest raw correlation, was confounded by domain size and did not survive either partial correlation or the null model. D4 persistent features added value over the simpler cycle rank baseline only in the null model test. These negative results are as informative as the positive ones: they constrain which structural features carry genuine governance-relevant signal.

The descriptors are computable from any lineage graph (dbt manifests, data catalog exports, code-derived dependency graphs) and require no access to the underlying data. They provide continuous structural measurement where traditional governance assessment methods provide periodic snapshots.

A Reproducibility

All code, anonymized data, and experiment scripts are archived at Zenodo[1] and on GitHub.[2] The repository includes:

- `src/governance_descriptors/`: Python package implementing D1–D4 and statistical utilities (permutation Spearman, Benjamini–Hochberg FDR, partial Spearman correlation).
- `experiments/`: Experiment scripts that reproduce all tables in this paper.
- `data/`: Anonymized dbt lineage data (HMAC-SHA256 hashed node identifiers, domain labels, governance metadata).

Dependencies: Python 3.11+, NetworkX 3.x, NumPy, SciPy, pandas, GUDHI (for persistent homology), scikit-learn (for Experiment 4).
Anonymization: All node identifiers, domain names, and schema names in the production dbt data were replaced with HMAC-SHA256 hashes prior to export. No organizational identifiers, table names, or column names are present in the released data.

[1] DOI: 10.5281/zenodo.20100000
[2] `https://github.com/mhdk1602/multiscale-governance-descriptors`

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